Mini Project Work Book

Third Year Electronics and Telecommunication Engineering

Year 2025 – 2026

Group/Project ID: TE-ETC-2025-26-01

Team Members:

1. Mr. Praful Uttam Avhad
2. Ms. Ghuge Pooja Vijay
3. Ms. Jamkar Kalyani Balu

Project Title: Gas and smoke detector

Project Guide: Prof. M. G. Kude

Area of the Project: Embedded Systems / IoT / Electronic Instrumentation

Late G. N. Sapkal College of Engineering

Department of Electronics and Telecommunication Engineering

Year 2025-2026

Late G. N. Sapkal College of Engineering

Vision of the Institute

- To be a leading educational institute that offer quality technical education and social consciousness to produce the competent engineers for well-being of society.

Mission of the Institute

- To become a premier institute transforming aspiring engineers through quality, skill-based technical education.
- To foster a social mindset by means of community involvement programs.
- To develop competent engineers with an ability to meet intellectual and career challenges.

Vision of the Department

- To nurture socially responsible technocrats in Electronics and telecommunication Engineering, capable of creating technology-driven solutions.

Mission of the Department

- To transform aspiring E&TC engineers through quality, skill-based education that builds strong technical competence and industry-oriented abilities.
- To foster a socially responsible mindset among students through community-oriented activities and technology-driven outreach programs.
- To develop competent E&TC engineers equipped with strong analytical abilities, technical expertise, and adaptability to meet intellectual, professional, and career challenges.

Department of Electronics and Telecommunication Engineering

Program Outcomes (UG)

Students are expected to know and be able to:

Sr. No.	PO	Domain	Description
1.	**PO1**	**Engineering knowledge**	Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
2.	**PO2**	**Problem analysis**	Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
3.	**PO3**	**Design / Development of Solutions**	Design creative solutions for complex engineering problems and design / develop systems / components / processes to meet identified needs with consideration for the public health and safety, whole- life cost, net zero carbon, culture, society and environment as required. (WK5)
4.	**PO4**	**Conduct Investigations of Complex Problems**	Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
5.	**PO5**	**Engineering Tool Usage**	Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
6.	**PO6**	**The Engineer and The World**	Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

7.	**PO7**	**Ethics**	Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
8.	**PO8**	**Individual and Team Work**	Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
9.	**PO9**	**Communication Skills**	Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
10.	**PO10**	**Project Management and Finance**	Apply knowledge and understanding of engineering management principles and economic decision- making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
11.	**PO11**	**Life-long Learning**	Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Educational Objectives (UG):

1. To prepare students with sound knowledge in mathematical, scientific and engineering fundamentals so as to enable them to have successful career in Electronics and Telecommunication Engineering (Pre-preparation).
2. To prepare students to adopt to emerging technologies in Electronics and Telecommunication Engineering (Core Competence).
3. To enable students to deploy their knowledge and skills in multidisciplinary fields (Breadth).
4. To inculcate professional attributes and ethics to work in team (Professionalism).

Program Specific Outcomes (PSO) (UG):

1. Understand analyze and develop fundamentals of electronics and Telecommunication in various domains of Analog and Digital systems.
2. Apply standard practices and strategies from the courses related to Microwave, Signal processing, Microcontroller, Embedded, networking and Communication Systems environment to deliver a quality product for business success.
3. Employ Hardware Skills, modern computer languages and platforms in creating innovative career paths to be an entrepreneur.

General Instructions

1. Students should enter the correct information in the work book.

2. Get all entries verified by respective project guide. No changes are to be made without project guide permission.

3. Students should report to their respective guides as per the schedule and its log is to be maintained in the work book.

4. Follow all deadlines and submit all documents strictly as per prescribed formats.

5. The work book should be produced at the time of all discussions, presentations and examinations.

6. The work book must be submitted to project coordinator/ guide/ department /College after successful examination at the end of year.

7. All documents and reports are to be prepared in Latex only (All the formats specifications provided adheres to MS Word but consequently applicable to final project report published using Latex)

8. Submit hard as well as soft copy. Maintain one copy with each member.

Savitribai Phule Pune University Third Year of E & Tc Engineering (2019 Course) 304200: Mini Project		
Teaching Scheme:	**Credit**	**Examination Scheme:**
Practical: 04 hrs. / week	**02**	**Term Work: 25 Marks** **Oral: 50 Marks**

Course Objectives:

- To understand the ―Product Development Process" including budgeting through Mini Project.
- To plan for various activities of the project and distribute the work amongst team members.
- To inculcate electronic hardware implementation skills by -
- Learning PCB artwork design using an appropriate EDA tool.
- Imbibing good soldering and effective trouble-shooting practices.
- Following correct grounding and shielding practices.
- To develop student's abilities to transmit technical information clearly and test the same by delivery of Seminar based on the Mini Project.
- To understand the importance of document design by compiling Technical Report on the Mini Project work carried out.

Course Outcomes:

On completion of the course, student will be able to
CO1: Understand, plan and execute a Mini Project with team.

CO2: Implement electronic hardware by learning PCB artwork design, soldering techniques, testing and troubleshooting etc.

CO3: Prepare a technical report based on the Mini project.

CO 4: Deliver technical seminar based on the Mini Project work carried out.

Guidelines for Selection of Project Work:

Project group shall consist of **not more than 3** students per group.

Mini Project Work should be carried out in the Design / Projects Laboratory.

Project designs ideas can be necessarily adapted from recent issues of electronic design magazines

Application notes from well-known device manufacturers may also be referred.

Use of Hardware devices/components is mandatory.

Layout versus schematic verification is mandatory.

Bare board test report shall be generated.

Assembly of components and enclosure design is mandatory.

Late G. N. Sapkal College of Engineering

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UNDERTAKING BY STUDENT

We, the students of T.E. E&T.C hereby assure that we will follow all the rules and Regulations related to project activity for the academic year 2025-2026. The Project entitled-
DIY Digital Oscilloscope Using Raspberry Pi Pico

Will be fully designed/ developed by us and every part of the project will be original work and will not be copied/ purchased from any source.

Name of the student	Signature
1. Mr. Avhad Praful Uttam	______________
2. Ms. Ghuge Pooja Vijay	______________
3. Ms. Jamkar Kalyani Balu	______________

Schedule of Project Work

Semester II

Sr. No.	Activity Scheduled	Date
1	Registration of Project groups	First week of Feb
2	Submission of Project Synopsis(Three topics)	Second Week of Feb
3	Project presentations	Third week of Feb
4	Finalization of projects & allotment of guide	Last week of Feb
5	Submission of final synopsis	first week of March
6	First presentation about progress of project work (Review I)	End of March
7	Technofest Competition	Last week of March
8	(Review II)	first week of April
9	Submission project report	Second of April
10	Project work Examination	As per SPPU Notification

Project Review (I)

Parameter	Related PO, PSO	Good (3)	Moderate (2)	Inadequate (1)
Selection of Topic	PO1			
Analysis and Synthesis	PO2,3,4			
Literature Survey	PO1, 2			
Ethical Attitude	PO8			
Independent Learning	PO9			
Oral Presentation	PO10			
Report Writing	PO10			
Continuous Learning	PO11			

Remarks :-

Name and Signature of Project Review Committee:

Prof. Archana Nimone
Guide and Project Coordinator

Prof. S. G. Bagul
HOD

Project Review (II)

Parameter	Related PO, PSO	Good (3)	Moderate (2)	Inadequate (1)
Analysis	PO2			
Design competence	PO3			
Problem Solving	PO4			
Usage of Techniques & Tools	PO5			
Project work impact on Society	PO6			
Project work impact on Environment	PO7			
Ethical attitude	PO8			
Independent Learning	PO9			
Oral Presentation	PO10			
Report Writing	PO10			
Time and Cost Analysis	PO11			
Continuous learning	PO11			

Internal Evaluation Sheet -TW

Sr No	Name of Student in Project Group	Problem statement /Objective/ Feasibility (05)	Literature Survey (05)	Requirement analysis Modelling and Design (5)	Presentation and Question Answer (5)	Project Report (5)	Total (25)
1	**Avhad Praful Uttam**						
2	**Ghuge Pooja Vijay**						
3	**Jamkar Kalyani Balu**						

Remarks :-

Name and Signature of Project Review Committee:
1. Prof. S. G. Bagul
2. Prof. S. D. Raul

Prof. Archana Nimone
Guide and Project Coordinator

Prof. S. G. Bagul
HOD